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Connection module

Abstract

A connection module includes a FPC, bus bars connected to the FPC and connecting the electrode terminals of power storage elements that are adjacent to each other, and a resin protector holding the bus bars and the FPC. The FPC includes a FPC body member and a first deformable portion that connects the FPC body member and the bus bars. The resin protector includes a FPC holding portion to which the FPC body member is fixed, movable bus bar holding portions to which the bus bars are fixed, respectively, and second deformable portions that connect the FPC holding portion and the movable bus bar holding portions. The first deformable portion and the second deformable portion connect the bus bars and the movable bus bar holding portions to the FPC body member and the FPC holding portion, respectively, such that the bus bars and the movable bus bar holding portions are movable with respect to the FPC body member and the FPC holding portion.

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Background/Summary

TECHNICAL FIELD

(1) The technology disclosed herein relates to a connection module.

BACKGROUND ART

(2) A connection module for an electric automobile or a hybrid automobile includes a battery block and a connection module. The battery block includes battery cells and the connection module is attached to the battery block to connect the battery cells. A flexible printed wiring board with bus bars has been known as such a connection module (refer to Patent Document 1). The flexible printed wiring board includes a flexible printed circuit board (FPC) and bus bars that are connected to the flexible printed circuit board and connect electrode terminals of adjacent power storage elements.

RELATED ART DOCUMENT

Patent Document

(3) [Patent Document 1] Japanese Unexamined Patent Application Publication No. 2014-86246

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

(4) The battery block including the battery cells has dimension tolerance due to manufacturing dimension errors among the battery cells and mounting errors of the battery cells. The dimension tolerance causes position displacement between the bus bar and the electrodes of the battery cells and this may hinder the attachment of the connection module to the battery block. The flexible printed wiring board having flexibility is not easy to be handled when being mounted on the battery block and this deteriorates mounting operability.

Means for Solving the Problem

(5) A connection module described herein is a connection module to be mounted on a power storage element group including power storage elements having electrode terminals and connecting the power storage elements. The connection module includes a flexible printed circuit board,

connection members (10) connected to the flexible printed circuit board and connecting the electrode terminals of the power storage elements that are adjacent to each other, and a holding member (40) holding the connection members and the flexible printed circuit board. The flexible printed circuit board includes a board body member and a first deformable portion (31) that connects the board body member and the connection members. The holding member includes a board holding portion (41) to which the board body member is fixed, connection member holding portions (61) to which the connection members are fixed, respectively, and a second deformable portion (51) that connects the board holding portion and the connection member holding portions. The first deformable portion and the second deformable portion connect the connection members (10) and the connection member holding portions (61) to the board body member (21) and the board holding portion (41), respectively, such that the connection members and the connection member holding portions are movable with respect to the board body member and the board holding portion.

(6) According to the above configuration, the connection members and the connection member holding portions that are connected to each other is movable with respect to the board body member and the board holding portion. Therefore, the difficulty of mounting the connection module on the power storage element group that may be caused by the position displacement of the electrode terminals due to the dimension tolerance of the power storage element group is less likely to be caused and this improves mounting operability. Since the flexible printed circuit board and the connection members are mounted on the holding member, the flexible printed circuit board having flexibility can maintain its shape and the flexible printed circuit board and the connecting members are collectively mounted in a predefined position on the power storage element group. This improves mounting operability.

(7) In the above configuration, the first deformable portion may extend from the board body member and include a first spring portion of an elongated plate spring having at least one curved portion.

(8) According to such a configuration, the connecting members can be connected to the board body member so as to be movable with respect to the board body member with a simple configuration. Since the first deformable portion is formed of the elongated plate spring, the connecting members are movable to some extent in each of the direction so as to be closer to and farther away from the board body member, the thickness direction of the board body member, and the direction along the board body member. Therefore, the flexible printed circuit board that is connected to the connecting members can be mounted on the holding member easily.

(9) In the above configuration, the second deformable portion may include second deformable portions that are arranged along one edge of the board holding portion. Each of the second deformable portions may include a second spring portion that can be expanded and contracted in a direction along the one edge.

(10) According to such a configuration, the connection member holding portions are allowed to move in the direction along one edge of the board holding portion by the second spring portion. The connection module is mounted on the power storage element group such that the one edge of the board holding portion is parallel to the arrangement direction of the power storage elements. With such a configuration, the connection member holding portions and the connection members that are held by the connection member holding portions are allowed to move in the arrangement direction of the power storage element group. Accordingly, difficulty of mounting the connection module on the power storage element group caused by the position displacement of the electrode terminals due to the dimension tolerance of the power storage element group is less likely to be caused and the mounting operability is improved.

(11) The holding member may include a connecting portion (81) connecting the connection member holding portions that are adjacent to each other so as to be movable in the direction along the one edge.

- (12) According to such a configuration, the adjacent connection member holding portions can be connected to each other and the connection members can be stably held by the connection member holding portions without hindering the movement of the connection member holding portions in the arrangement direction of the power storage element group.
- (13) In the above configuration, the board holding portion (41) may include a holding body portion (42) that extends along the board body member, and a retaining piece that is spaced from the holding body portion and holding the board body member with the holding body portion.
- (14) According to such a configuration, the board body member can be held by the board holding portion with a simple structure.
- (15) In the above configuration, the board body member (21) may include a positioning hole, and the board holding portion may include a holding body portion (42) that extends along the board body member and a positioning projection that projects from the holding body portion and is inserted in the positioning hole to position the board body portion with respect to the holding body portion.
- (16) According to such a configuration, the board body member can be positioned on the board holding portion with a simple structure.
- (17) In the above configuration, each of the connection member holding portions (61) may include a base portion and a stopper that is continuous from the base portion. The stopper may include a warping portion that projects from the base portion and can be warped and a stopper projection that projects from the warping portion and sandwiches and holds one of the connection members (10) with the base portion.
- (18) According to such a configuration, the connection members are easily mounted on the connection member holding portions, respectively, only by pushing the connection members toward the base portion. This improves assembling operability of the connection module.
- Advantageous Effects of Invention
- (19) According to the connection module described herein, mounting operability can be improved.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a power storage module according to one embodiment.
- (2) FIG. 2 is an upper view of the power storage module according to the embodiment.
- (3) FIG. 3 is a perspective view of a connection module according to the embodiment.
- (4) FIG. 4 is an upper view of the connection module according to the embodiment.
- (5) FIG. 5 is an enlarged view illustrating a portion surrounded by a frame R1 in FIG. 4.
- (6) FIG. 6 is a cross-sectional view taken along A-A line in FIG. 5.
- (7) FIG. 7 is a cross-sectional view taken along B-B line in FIG. 5.
- (8) FIG. 8 is a cross-sectional view taken along C-C line in FIG. 5.
- (9) FIG. 9 is a perspective view of a bus bar according to the embodiment.
- (10) FIG. 10 is an upper view of the bus bar according to the embodiment.
- (11) FIG. 11 is an upper view of a flexible printed circuit board according to the embodiment.
- (12) FIG. 12 is a perspective view of a resin protector according to the embodiment.
- (13) FIG. 13 is an upper view of the resin protector according to the embodiment.
- (14) FIG. 14 is a front view of the resin protector according to the embodiment.
- (15) FIG. 15 is a rear view of the resin protector according to the embodiment.
- (16) FIG. 16 is an enlarged view illustrating a portion surrounded by a frame R2 in FIG. 12.

MODES FOR CARRYING OUT THE INVENTION

- (17) One embodiment will be described with reference to FIGS. 1 to 16. A connection module 1 according to this embodiment is included in a power storage module M that is used as a power

source for driving a vehicle such as an electric automobile or a hybrid automobile. As illustrated in FIG. 1, the connection module 1 is attached to a power storage element group 90G including power storage elements 90 that are arranged in a row to connect the power storage elements 90 in series.

(18) [Power Storage Element 90 and Power Storage Element Group 90G]

(19) The power storage element 90 is a secondary battery, for example. As illustrated in FIG. 1, each of the power storage elements 90 has a rectangular parallelepiped flat shape and has an electrode mount surface 90F (an upper surface in FIG. 1) that is vertical to a surface of the power storage element 90 opposite an adjacent power storage element 90. Electrode terminals 91A, 91B are disposed on the electrode mount surface 90F. One of the electrode terminals 91A, 91B is an anode terminal 91A and another one is a cathode terminal 91B. Each of the electrode terminals 91A, 91B has a columnar shape and has threads on a peripheral surface thereof.

(20) As illustrated in FIG. 1, the power storage elements 90 are arranged in a row and are configured as the power storage element group 90G. Among the power storage elements 90, every two adjacent power storage elements 90 are arranged such that the electrode terminals 91A, 91B having different polarities are disposed adjacent to each other (the anode terminal 91A of one power storage element 90 is disposed adjacent to the cathode terminal 91B of another power storage element 90 that is adjacent to the one power storage element 90).

(21) In the following description, an arrangement direction in which the power storage elements 90 are arranged (a lower-left to upper-right direction in FIG. 1) corresponds to an X-axis direction, a direction extending along a surface of one power storage element 90 that is opposite another power storage element 90 next to the one power storage element 90 (a lower-right to an upper-left direction in FIG. 1) corresponds to a Y-axis direction, and a direction vertical to the electrode mount surface 90F (an upper-lower direction in FIG. 1) corresponds to a Z-axis direction.

(22) [Connection Module 1]

(23) The connection module 1 is mounted on a surface (an upper surface in FIG. 1) of the power storage element group 90G including the electrode mount surfaces 90F of the power storage elements 90. As illustrated in FIG. 3, the connection module 1 includes a flexible printed circuit board 20 (hereinafter, referred to as a FPC 20), bus bars 10 (corresponding to a connection member), and a resin protector (corresponding to a holding member). The bus bars 10 are connected to the FPC 20 and each of the bus bars 10 connects the anode terminal 91A and the cathode terminal 91B of the adjacent power storage elements 90. The resin protector 40 holds the bus bars 10 and the FPC 20. In FIG. 1, only the connection module 1 that is connected to one of the two rows of the electrode terminals 91A, 91B (the row on the lower-right side) is illustrated; however, another connection module 1 is connected to the other one of the two rows. Further, in the connection module 1 in FIG. 1 and FIG. 2, one bus bar 10 connecting the anode terminal 91A at the right end and the cathode terminal 91B adjacent to the right-end anode terminal 91A is illustrated and other bus bars 10 are not illustrated.

(24) [Bus Bar 10]

(25) Each of the bus bars 10 is made of metal and includes an electrode connection portion 11, a FPC connection portion 15, and a stopper wall 16, as illustrated in FIGS. 9 and 10. The electrode connection portion 11 connects the anode terminal 91A and the cathode terminal 91B of the adjacent power storage elements 90. The FPC connection portion 15 is continuous from the electrode connection portion 11 and connected to the FPC 20. The stopper wall 16 is continuous from the FPC connection portion 15.

(26) The electrode connection portion 11 has a rectangular plate shape as a whole and has two electrode insertion holes 12 through which the electrode terminals 91A, 91B are inserted, respectively, and two recessed portions 13 to be connected to the resin protector 40. The electrode connection portion 11 includes one of the electrode insertion holes 12 close to one of short sides 11S thereof and another one close to another one of the short sides 11S. One of the two recessed portions 13 is recessed from one of the short sides 11S of the electrode connection portion 11 and

another one is recessed from the other one of the short sides **11S**.

(27) The electrode connection portion **11** has two long sides **11LA**, **11LB** and a connection recess **14** recessed from the long side **11LA**. The connection recess **14** is defined by a first edge **14A** that is parallel to the long side **11LA** and two first side edges **14B** that connect ends of the first edge **14A** to the long side **11LA**. The FPC connection portion **15** is a quadrangular plate portion that extends from the first edge **14A** along a same plane surface as the electrode connection portion **11**. The stopper wall **16** is a short wall plate portion that extends vertically from a distal end of the FPC connection portion **15**.

(28) [FPC **20**]

(29) The FPC **20** is for electrically connecting the bus bars **10** and an ECU (electric control unit, not illustrated) and is not illustrated in detail. The FPC **20** includes conductive wirings made of a copper foil and an insulating resin film that covers both surfaces of the conductive wirings. As illustrated in FIG. **11**, the FPC **20** includes a FPC body member **21** (corresponding to a base board), first deformable portions **31**, and connection portions **35**. The FPC body member **21** has a rectangular belt shape as a whole. The first deformable portions **31** are continuous from the FPC body member **21**. The connection portions **35** are continuous from the respective first deformable portions **31** and connected to the respective bus bars **10**.

(30) The FPC body member **21** includes a first slit **22**, a notch portion **23**, positioning holes **24**, and two fitting holes **25**.

(31) The FPC body member **21** includes the first slit **22** in a substantially middle section between the long sides **21LA**, **21LB** and the first slit **22** extends parallel to the long sides **21LA**, **21LB** over a substantially entire length of the FPC body member **21** except for two end portions thereof (near the short sides **21SA**, **21SB**). The notch portion **23** has a relatively wide opening space that extends from the long side **21LA** (a lower long side in FIG. **11**) of the FPC body member **21** to the first slit **22** and is closer to the short side **21SA** (on a right side in FIG. **11**). A portion of the FPC body member **21** sandwiched between the first slit **22** and the long side **21LA** (a portion lower than the first slit **22** in FIG. **11**) is divided into a first divided portion **26** and a second divided portion **27** by the notch portion **23**. The first divided portion **26** is long and on the short side **21SB** side with respect to the notch portion **23** and the second divided portion **27** is short and on the short side **21SA** side with respect to the notch portion **23**.

(32) One positioning hole **24** is adjacent to the short side **21SA**, two positioning holes **24** are adjacent to the short side **21SB**, and the other one is at a substantially middle of the first divided portion **26**. One of the two fitting holes **25** is included in the first divided portion **26** and close to the notch portion **23** and another one of the two fitting holes **25** is included in the second divided portion **27** and close to the notch portion **23**.

(33) The first deformable portion **31** is an elongated plate spring portion (corresponding to a first spring portion) that is continuous from the FPC body member **21**. The first deformable portion **31** includes a first curved portion **32** (corresponding to a curved portion), a straight portion **33**, and a second curved portion **34** (corresponding to the curved portion). The first curved portion **32** extends from the long side **21LA** of the FPC body member **21** and is curved into a U-shape. The straight portion **33** extends from an extended end of the first curved portion **32** along the long side **21LA**. The second curved portion **34** extends from an extended end of the straight portion **33** and is curved into a U-shape in an opposite direction from the first curved portion **32**. Thus, the first deformable portion **31** has a substantially S-shape as a whole. The first deformable portions **31** are arranged in a row along the long side **21LA** and one of the first deformable portions **31** extends from the second divided portion **27** and the rest of them extend from the first divided portion **26**.

(34) The connection portion **35** extends from the extended end of the second curved portion **34** and is a quadrangular plate portion. A portion of the wirings is exposed on a surface of the connection portion **35** as a connection land (not illustrated) and the FPC connection portion **15** is connected to the connection land with soldering.

(35) [Resin Protector **40**]

(36) The resin protector **40** is made of synthetic resin and includes a FPC holding portion **41** (corresponding to a board holding portion) that holds the FPC body member **21** and bus bar holding portions **61**, **71** that hold the bus bars **10**.

(37) As illustrated in FIG. **12**, the FPC holding portion **41** includes a holding plate **42** (corresponding to a holding body portion), a side rib **43** and center ribs **44A**, **44B** that extend from the holding plate **42**, retaining pieces **45**, positioning projections **46**, and two FPC stopper pieces **47**.

(38) The holding plate **42** is a rectangular plate that has a substantially same size as that of the FPC body member **21** as a whole. The holding plate **42** has one surface (an upper surface in FIG. **12**) as a mount surface **42F** on which the FPC body member **21** is disposed and includes the side rib **43**, the center ribs **44A**, **44B**, the retaining pieces **45**, the positioning projections **46**, and the FPC stopper pieces **47** on the mount surface **42F**.

(39) The side rib **43** is a thin elongated portion that projects from the long side **42LA** (the upper one in FIG. **12**) out of the long sides **42LA**, **42LB** of the holding plate **42** and extends over a substantially entire length of the long side **42LA**. The center ribs **44A**, **44B** are thin elongated portions that extend parallel to the long sides **42LA**, **42LB** at a substantially middle between the long sides **42LA**, **42LB**. The long center rib is disposed adjacent to the short side **42SA** (on the left side in FIG. **12**) out of the short sides **42SA**, **42SB** of the holding plate **42** and the short center rib is disposed adjacent to the short side **42SB**.

(40) Each of the retaining pieces **45** is a plate member that extends from the side rib **43** parallel to the holding plate **42** and retains and sandwiches the FPC body member **21** with the holding plate **42**. Each of the positioning projections **46** is a circular cylindrical projection projecting from the holding plate **42** to correspond to each of the positioning holes **24** in the FPC body member **21**. As illustrated in FIG. **8**, each of the two FPC stopper pieces **47** includes a stopper piece body **47A** and a stopper projection **47B** and is disposed to correspond to each of the two fitting holes **25** in the FPC body member **21**. The stopper piece body **47A** extends from the holding plate **42** and the stopper projection **47B** projects from a distal end of the stopper piece body **47A**.

(41) The holding plate **42** includes recessed portions **48** for springs that are recessed inward from the long side **42LB** (corresponding to one edge). As illustrated in FIG. **16**, the recessed portion **48** is defined by a second edge **48A** that is parallel to the long side **42LB** and two second side edges **48B** that connect ends of the second edge **48A** to the long side **42LB**, respectively.

(42) As illustrated in FIG. **12**, one of the bus bar holding portions **61**, **71** closest to the short side **42SA** of the holding plate **42** is the fixed bus bar holding portion **71** and other ones are movable bus bar holding portions **61** (correspond to a connection member holding portion). The movable bus bar holding portions **61** are connected to the holding plate **42** via second deformable portions **51**, respectively.

(43) As illustrated in FIGS. **15** and **16**, the second deformable portion **51** is a plate member that is folded in a bellows form as a whole so as to be expanded and contracted in a direction along the long side **42LB** of the holding plate **42**. More specifically, the second deformable portion **51** includes a spring connection portion **52** (corresponding to a spring connection portion) and two spring plate portions **53** (corresponding to a second spring portion). The spring connection portion **52** has a plate shape extending from the second edge **48A** along a same plane surface as the holding plate **42**. The spring plate portions **53** extend from the spring connection portion **52** along the second edge **48A** in opposite directions while being bent. Each of the spring plate portions **53** is an S-shaped plate spring. Each of the spring plate portions **53** extends vertically from each of the extended ends of the spring connection portion **52** so as to be away from the holding plate **42** and is folded back and extends toward the holding plate **42** and is folded back again and extends away from the holding plate **42**.

(44) As illustrated in FIG. **16**, the movable bus bar holding portion **61** includes a back plate **62** that

is continuous from the second deformable portion 51, a bottom plate portion 63 (corresponding to the base portion) that is continuous from the back plate 62, an extended portion 65 extending from the bottom plate portion 63, a first bus bar stopper 66 (corresponding to a stopper), two second bus bar stoppers 67 (corresponding to the stopper), and two contact plates 68.

(45) As illustrated in FIG. 16, the back plate 62 is a plate that is vertical to the holding plate 42 and is connected to each of the distal ends of the spring plate portions 53.

(46) As illustrated in FIG. 16, the bottom plate portion 63 is a plate that extends vertically from the back plate 62 in an opposite direction from the holding plate 42 and includes two second slits 64. As illustrated in FIGS. 13 and 16, each of the two second slits 64 extends from the extended edge of the bottom plate portion 63 toward the back plate 62. The bottom plate portion 63 is divided into two edge plate portions 63A at two ends thereof and a middle plate portion 63B at a middle by the second slits 64. The extended portion 65 is a plate portion that extends from the extended edge of the bottom plate portion 63 along a same plane surface as that of the bottom plate portion 63.

(47) As illustrated in FIGS. 6 and 16, the first bus bar stopper 66 includes a first warping portion 66A (corresponding to a warping portion) and a first stopper projection 66B (corresponding to a stopper projection). The first warping portion 66A extends from the middle plate portion 63B and is spaced away from the back plate 62. The first stopper projection 66B projects from the extended end of the first warping portion 66A in a direction opposite from the back plate 62. The first warping portion 66A is slightly tilted so as to be farther away from the back plate 62 as it extends from the middle plate portion 63B.

(48) As illustrated in FIG. 7, each of the two second bus bar stoppers 67 includes a second warping portion 67A (corresponding to the warping portion) and a second stopper projection 67B (corresponding to the stopper projection). The second warping portions 67A extend vertically from the extended edges of the two edge plate portions 63A, respectively and the second stopper projections 67B project from the distal ends of the second warping portions 67A toward the back plate 62, respectively.

(49) As illustrated in FIG. 16, each of the two contact plates 68 is a plate portion that extends from the middle plate portion 63B along a slit edge of the corresponding one of the two second slits 64. The contact plates 68 are disposed next to the back plate 62.

(50) Each of the movable bus bar holding portions 61 is allowed to move to some extent with respect to the holding plate 42 by the second deformable portion 51. Specifically, the movable bus bar holding portion 61 is movable in a direction along the long side 42LB (the X-axis direction) due to expansion and contraction of the two spring plate portions 53 of the second deformable portion 51.

(51) As illustrated in FIG. 12, the fixed bus bar holding portion 71 does not have the second deformable portion 51 but has a configuration similar to that of the movable bus bar holding portion 61. The fixed bus bar holding portion 71 includes a back plate 72 extending from the long side 42LB of the holding plate 42. The components of the fixed bus bar holding portion 71 same as those of the movable bus bar holding portion 61 are represented by the same symbols as those of the movable bus bar holding portion 61 and will not be described.

(52) The fixed bus bar holding portion 71 is connected to the adjacent movable bus bar holding portion 61 that is adjacent to the fixed bus bar holding portion 71 by a connecting portion 81 of a U-shaped plate spring and the adjacent movable bus bar holding portions 61 are connected to each other by the connecting portions 81. Accordingly, the fixed bus bar holding portion 71 is connected to the adjacent movable bus bar holding portion 61 and the adjacent movable bus bar holding portions 61 are connected to each other such that the movable bus bar holding portions 61 can move in the direction along the long side 42LB of the holding plate 42 (the X-axis direction). Further, the bus bars 10 can be stably held by the fixed bus bar holding portion 71 and the movable bus bar holding portions 61.

(53) [Assembling of Connection Module 1]

(54) One example of steps of assembling the connection module **1** having the above configuration will be described below.

(55) First, the bus bars **10** are connected to the FPC **20**. The FPC connection portions **15** of the respective bus bars **10** are put on the respective connection portions **35** of the FPC **20** and the FPC connection portions **15** and the connection portions **35** are connected to each other, respectively, with reflow soldering. As illustrated in FIGS. **3** and **5**, the bus bars **10** that are connected to the FPC **20** are connected to the FPC body member **21** via the first deformable portions **31**. Accordingly, the bus bars **10** can freely move to some extent in each of the direction along the long side **21LA** of the FPC body member **21** (the X-axis direction), the direction to be closer to and farther away from the FPC body member **21** (the Y-axis direction), and the thickness direction of the FPC body member **21** (the Z-axis direction) by the deformation of the first deformable portions **31**.

(56) Next, the FPC **20** connected to the bus bars **10** is mounted on the resin protector **40**.

(57) First, the FPC body member **21** is mounted on the FPC holding portion **41**. As illustrated in FIG. **3**, the FPC body member **21** is inserted into spaces between the holding plate **42** and the retaining pieces **45** to be placed on the holding plate **42** such that the long side **21LB** is along the side rib **43**. The positioning projections **46** are inserted in the positioning holes **24**, respectively, and the two center ribs **44A**, **44B** are fitted in the first slit **22** and thus, the FPC body member **21** is positioned on the holding plate **42**. The retaining pieces **45** retain the FPC body member **21** such that the FPC body member **21** is not detached from the holding plate **42**. As illustrated in FIGS. **3** and **8**, the stopper piece bodies **47A** of the FPC stopper pieces **47** are inserted in the fitting holes **25**, respectively, and the stopper projections **47B** are contacted with the FPC body member **21** to retain the FPC body member **21**. Thus, the edge portions of the respective divided portions **26**, **27** are less likely to be peeled off from the holding plate **42**.

(58) Next, the bus bars **10** are mounted on the bus bar holding portions **61**, **71**, respectively. Each second bus bar stopper **67** is inserted in each recessed portion **13** and the electrode connection portion **11** is pushed toward the bottom plate portion **63** while the first bus bar stopper **66** being warped by the stopper wall **16**. When the electrode connection portion **11** is contacted with the bottom plate portion **63**, as illustrated in FIG. **6**, the first bus bar stopper **66** elastically restores its original shape and the stopper wall **16** is sandwiched between the middle plate portion **63B** and the first stopper projection **66B**. As illustrated in FIGS. **5** and **7**, the portion of the electrode connection portion **11** around the recessed portion **13** is disposed between the bottom plate portion **63** and the second stopper projection **67B**. Thus, the bus bars **10** are fixed to the bus bar holding portions **61**, **71**, respectively. In such a configuration, the bus bars **10** are allowed to move freely to some extent with respect to the FPC body member **21** since the first deformable portions **31** can be deformed. Therefore, the bus bars **10** can be mounted on the bus bar holding portions **61**, **71**, respectively, easily. The bus bars **10** are easily mounted on the bus bar holding portions **61**, **71**, respectively, only by pushing the bus bars **10** toward the bottom plate portion **63**. This improves assembling operability of the connection module **1**.

(59) As illustrated in FIG. **3**, when the bus bars **10** are mounted on the bus bar holding portions **61**, **71**, respectively, the first curved portion **32** is disposed on the same plane surface as that of the FPC body member **21** and the straight portion **33** is tilted to be farther away from the FPC body member **21** as it extends farther away from the first curved portion **32**. The second curved portion **34**, the connection portion **35**, and the bus bars **10** are disposed parallel to the FPC body member **21** and at a level different from the level of the FPC body member **21**. The straight portion **33** is disposed in a space between the first bus bar stopper **66** and the back plate **62**.

(60) In the connection module **1**, the FPC body member **21** is held by and positioned with respect to the FPC holding portion **41**. The bus bars **10** are held by and positioned with respect to the movable bus bar holding portions **61**, respectively. The bus bars **10** are movably connected to the FPC body member **21** via the first deformable portions **31**, respectively, and the movable bus bar holding portions **61** are movably connected to the FPC holding portion **41** via the second

deformable portions **51**, respectively. According to such a configuration, the movable bus bar holding portions **61** and the bus bars **10** that are connected to each other, respectively, are allowed to move with respect to the FPC holding portion **41** and the FPC body member **21** in the direction along the long side **42LB** of the holding plate **42** and the long side **21LA** of the FPC body member **21** (the X-axis direction).

(61) [Mounting of Connection Module **1** on Power Storage Element Group **90G**]

(62) One example of steps of mounting the connection module **1** having the above configuration on the power storage element group **90G** will be described below.

(63) As illustrated in FIGS. **1** and **2**, the connection module **1** is disposed in a predefined position on the power storage element group **90G** and the electrode terminals **91A**, **91B** are inserted in the electrode insertion holes **12** of the bus bars **10**, respectively. Then, nuts, which are not illustrated, are screwed on the respective electrode terminals **91A**, **91B** to connect the electrode terminals **91A**, **91B** and each of the bus bars **10**.

(64) In the power storage element group **90G** including the power storage elements **90**, dimension tolerance may be caused due to manufacturing dimension errors in the power storage elements **90** and mounting errors of the power storage elements **90**. The dimension tolerance may cause position displacement in the electrode terminals **91A**, **91B** in the arrangement direction of the power storage elements **90** (the X-axis direction). In the present embodiment, as previously described, the movable bus bar holding portions **61** and the bus bars **10** are movable with respect to the FPC holding portion **41** and the FPC body member **21**. Therefore, each of the bus bars **10** can be moved corresponding to the position displacement of the electrode terminals **91A**, **91B** caused by the dimension tolerance in the power storage element group **90G** so as to be mounted on the electrode terminals **91A**, **91B**.

(65) For example, if the distance between the electrode terminals **91A**, **91B** is smaller than the predefined design dimension, the second deformable portions **51** and the connecting portions **81** are deformed to move the movable bus bar holding portions **61** to be closer to the fixed bus bar holding portion **71** (in the direction along the long side **42LB** of the holding plate **42**, in a direction from an upper right side toward a lower left side in FIG. **1**) that is referred to as a reference position. In other words, the movable bus bar holding portions **61** are moved with reference to the fixed bus bar holding portion **71** that is disposed at an end of the resin protector **40** such that a distance between the fixed bus bar holding portion **71** and the movable bus bar holding portion **61** that is next to the fixed bus bar holding portion **71** and distances between the adjacent movable bus bar holding portions **61** become smaller. Accordingly, with reference to one bus bar **10** held on the fixed bus bar holding portion **71**, other bus bars **10** are moved such that the distances between the adjacent bus bars **10** become smaller to deal with the position displacement of the electrode terminals **91A**, **91B**.

(66) If the distance between the electrode terminals **91A**, **91B** is greater than the predefined design dimension, the second deformable portions **51** and the connecting portions **81** are deformed to move the movable bus bar holding portions **61** to be farther away from the fixed bus bar holding portion **71** (in the direction along the long side **42LB** of the holding plate **42**, in a direction from the lower left side toward the upper right side in FIG. **1**) that is referred to as the reference position. In other words, the movable bus bar holding portions **61** are moved with reference to the fixed bus bar holding portion **71** that is disposed at an end of the resin protector **40** such that the distance between the fixed bus bar holding portion **71** and the movable bus bar holding portion **61** that is next to the fixed bus bar holding portion **71** and the distances between the adjacent movable bus bar holding portions **61** become greater. Accordingly, with reference to one bus bar **10** held on the fixed bus bar holding portion **71**, other bus bars **10** are moved such that the distances between the adjacent bus bars **10** become greater to deal with the position displacement of the electrode terminals **91A**, **91B**.

(67) Accordingly, difficulty of mounting the connection module **1** on the power storage element

group **90G** caused by the dimension tolerance of the power storage element group **90G** is less likely to be caused and the mounting operability is improved.

(68) Since the FPC **20** and the bus bars **10** are mounted on the resin protector **40**, the FPC **20** having flexibility can maintain its shape and the FPC **20** and the bus bars **10** are collectively mounted in a predefined position on the power storage element group **90G**. This improves mounting operability.

(69) [Summary]

(70) As previously described, according to the present embodiment, the connection module **1** is a module that is mounted on the power storage element group **90G** including the power storage elements **90** having the electrode terminals **91A**, **91B** and connects the power storage elements **90**. The connection module **1** includes the FPC **20**, the bus bars **10**, and the resin protector **40**. The bus bars **10** are connected to the FPC **20** and connect the electrode terminals **91A**, **91B** of the adjacent power storage elements **90**. The resin protector **40** holds the bus bars **10** and the FPC **20**. The FPC **20** includes the FPC body member **21** and the first deformable portions **31** that connect the FPC body member **21** and the bus bars **10**. The resin protector **40** includes the FPC holding portion **41** on which the FPC body member **21** is fixed, the movable bus bar holding portions **61** on which the respective bus bars **10** are fixed, and the second deformable portions **51** that connect the FPC holding portion **41** and the respective movable bus bar holding portions **61**. The first deformable portions **31** and the second deformable portions **51** connect the bus bars **10** and the movable bus bar holding portions **61** to the FPC body member **21** and the FPC holding portion **41** such that the bus bars **10** and the movable bus bar holding portions **61** are movable with respect to the FPC body member **21** and the FPC holding portions **41**.

(71) According to the above configuration, the bus bars **10** and the movable bus bar holding portions **61** that are connected to each other is movable with respect to the FPC body member **21** and the FPC holding portion **41**. Therefore, the difficulty of mounting the connection module **1** on the power storage element group **90G** that may be caused by the position displacement of the electrode terminals **91A**, **91B** due to the dimension tolerance of the power storage element group **90G** is less likely to be caused and this improves mounting operability. Since the FPC **20** and the bus bars **10** are mounted on the resin protector **40**, the FPC **20** having flexibility can maintain its shape and the FPC **20** and the bus bars **10** are collectively mounted in a predefined position on the power storage element group **90G**. This improves mounting operability.

(72) The first deformable portion **31** is an elongated plate spring that extends from the FPC body member **21** and includes the first curved portion **32** and the second curved portion **34**.

(73) According to such a configuration, the bus bars **10** can be connected to the FPC body member **21** so as to be movable with respect to the FPC body member **21** with a simple configuration. Since the first deformable portions **31** are formed of the elongated plate springs, the bus bars **10** are movable to some extent in each of the direction so as to be closer to and farther away from the FPC body member **21**, the thickness direction of the FPC body member **21**, and the direction along the FPC body member **21**. Therefore, the FPC **20** that is connected to the bus bars **10** can be mounted on the resin protector **40** easily.

(74) The second deformable portions **51** are arranged along the long side **42LB** of the FPC holding portion **41** and each of the second deformable portions **51** includes the spring plate portion **53** that can be expanded and contracted in the direction along the long side **42LB**.

(75) According to such a configuration, the movable bus bar holding portions **61** are allowed to move in the direction along the long side **42LB** of the FPC holding portion **41** by the second deformable portions **51**. The connection module **1** is mounted on the power storage element group **90G** such that the long side **42LB** of the FPC holding portion **41** is parallel to the arrangement direction of the power storage elements **90**. With such a configuration, the movable bus bar holding portions **61** and the bus bars **10** that are held by the movable bus bar holding portions **61** are allowed to move in the arrangement direction of the power storage element group **90G**.

Accordingly, difficulty of mounting the connection module **1** on the power storage element group **90G** caused by the position displacement of the electrode terminals **91A**, **91B** due to the dimension tolerance of the power storage element group **90G** is less likely to be caused and the mounting operability is improved.

(76) The resin protector **40** may include the connecting portions **81** that connect the adjacent movable bus bar holding portions **61** to be movable in the direction along the holding plate **42** of the FPC holding portion **41**.

(77) According to such a configuration, the adjacent movable bus bar holding portions **61** can be connected to each other and the bus bars **10** can be stably held by the movable bus bar holding portions **61** without hindering the movement of the movable bus bar holding portions **61** in the arrangement direction of the power storage element group **90G**.

(78) The FPC holding portion **41** includes the holding plate **42** that is along the FPC body member **21** and the retaining pieces **45** that retain the FPC body member **21** with the holding plate **42**.

(79) According to such a configuration, the FPC body member **21** can be held by the FPC holding portion **41** with a simple structure.

(80) The FPC body member **21** includes the positioning holes **24**. The FPC holding portion **41** includes the holding plate **42** that is disposed along the FPC body member **21** and the positioning projections **46** that project from the holding plate **42** and are inserted in the positioning holes **24** to position the FPC body member **21** with respect to the holding plate **42**.

(81) According to such a configuration, the FPC body member **21** can be positioned on the FPC holding portion **41** with a simple structure.

(82) The movable bus bar holding portion **61** includes the bottom plate portion **63** that is along the bus bar **10**, and the first bus bar stopper **66** and the second bus bar stoppers **67** that are continuous from the bottom plate portion **63**. The first bus bar stopper **66** projects from the bottom plate portion **63** and includes the first warping portion **66A** that can be warped and the first stopper projection **66B** that projects from the first warping portion **66A** and sandwiches and holds the bus bar **10** with the bottom plate portion **63**. Each of the second bus bar stoppers **67** projects from the bottom plate portion **63** and includes the second warping portion **67A** that can be warped and the second stopper projection **67B** that projects from the second warping portion **67A** and sandwiches and holds the bus bar **10** with the bottom plate portion **63**.

(83) According to such a configuration, the bus bars **10** are easily mounted on the movable bus bar holding portions **61**, respectively, only by pushing the bus bars **10** toward the bottom plate portion **63**. This improves assembling operability of the connection module **1**.

Other Embodiments

(84) The technology disclosed herein is not limited to the embodiment described above and illustrated in the drawings. For example, the following embodiments will be included in the technical scope of the technology.

(85) (1) In the above embodiment, the first deformable portion **31** includes the first curved portion **32** and the second curved portion **34**; however, the deformable portion may include one curved portion, three or more curved portions.

(86) (2) In the above embodiment, the second deformable portion **51** is folded in a bellows form and includes the spring plate portions **53**; however, the shape of the second spring portion is not limited to the above but may be any form as long as the spring portion can be expanded and contracted in the direction along one edge of the base board holding portion.

(87) (3) In the above embodiment, the two spring plate portions **53** of the second deformable portion **51** have a symmetrical shape with respect to the spring connection portion **52**; however, two second spring portions may have an asymmetrical shape.

EXPLANATION OF SYMBOLS

(88) **1**: Connection module **10**: Bus bar (connection member) **20**: Flexible printed circuit board **21**: FPC body member (board body member) **24**: Positioning hole **31**: First deformable portion (first

spring portion) **32**: First curved portion (curved portion) **34**: Second curved portion (curved portion) **40**: Resin protector (holding member) **41**: FPC holding portion (board holding portion) **42**: Holding plate (holding body portion) **45**: Retaining piece **46**: Positioning projection **51**: Second deformable portion **53**: Spring plate portion (second spring portion) **61**: Movable bus bar holding portion (connection member holding portion) **63**: Bottom plate portion (base portion) **66**: First bus bar stopper (stopper) **66A**: First warping portion (warping portion) **66B**: First stopper projection (stopper projection) **67**: Second bus bar stopper (stopper) **67A**: second warping portion (warping portion) **67B**: Second stopper projection (stopper projection) **81**: Connecting portion **90**: Power storage element **90G**: Power storage element group **91A**, **91B**: Electrode terminal

Claims

1. A flexible printed circuit board to be connected to connection members that connect adjacent electrode terminals of power storage elements, the flexible printed circuit board comprising: a board body member; and first deformable portions that connect the board body member and the connection members such that the connection members are movable with respect to the board body member, wherein each of the first deformable portions includes: a first curved portion that extends from the board body member and is curved into a U-shape, a straight portion that extends from an extended end of the first curved portion along one side edge of the board body member, and a second curved portion that extends from an extended end of the straight portion and is curved into a U-shape in an opposite direction from the first curved portion, and each of the first deformable portions has an S-shape as a whole, and the first curved portion that extends from the board body member includes a proximal portion that is recessed with respect to the board body member, such that an endmost portion of the board body member, in a direction orthogonal to an arrangement direction of the power storage elements, extends further than the proximal portion of the first curved portion.
2. The flexible printed circuit board according to claim 1, wherein the first deformable portions extend from the board body member and each of the first deformable portions includes a first spring portion that is an elongated plate spring and includes at least the first curved portion.
3. The flexible printed circuit board according to claim 1, wherein the board body member includes recessed portions that are recessed from an outer edge of the board body member, and the first deformable portions extend from inner edges of the recessed portions.
4. The flexible printed circuit board according to claim 1, wherein the board body member includes a positioning hole.
5. A connection module that is to be mounted on a power storage element group including power storage elements having electrode terminals and connects the power storage elements, the connection module comprising: a flexible printed circuit board; connection members connected to the flexible printed circuit board and connecting the electrode terminals of the power storage elements that are adjacent to each other; and a holding member holding the connection members and the flexible printed circuit board, wherein the flexible printed circuit board includes a board body member and first deformable portions that connect the board body member and the connection members, and the first deformable portions connect the board body member and the connection members such that the connection members are movable with respect to the board body member, each of the first deformable portions includes: a first curved portion that extends from the board body member and is curved into a U-shape, a straight portion that extends from an extended end of the first curved portion along one side edge of the board body member, and a second curved portion that extends from an extended end of the straight portion and is curved into a U-shape in an opposite direction from the first curved portion, and each of the first deformable portions has an S-shape as a whole, and the first curved portion that extends from the board body member includes a proximal portion that is recessed with respect to the board body member, such that an endmost

- portion of the board body member, in a direction orthogonal to an arrangement direction of the power storage elements, extends further than the proximal portion of the first curved portion.
6. The connection module according to claim 5, wherein the first deformable portions extend from the board body member and each of the first deformable portions includes a first spring portion that is an elongated plate spring and includes at least the first curved portion.
7. The connection module according to claim 5, wherein the holding member includes a board holding portion to which the board body member is fixed, connection member holding portions to which the connection members are fixed, and second deformable portions that connect the board holding portion and the connection member holding portions, and the second deformable portions connect the connection member holding portions and the board holding portion such that the connection member holding portions are movable with respect to the board holding portion.
8. The connection module according to claim 7, wherein the second deformable portions are arranged along one edge of the board holding portion, and each of the second deformable portions includes a second spring portion that can be expanded and contracted in a direction along the one edge of the board holding portion.
9. The connection module according to claim 8, wherein the holding member includes a connecting portion connecting the connection member holding portions that are adjacent to each other so as to be movable in the direction along the one edge of the board holding portion.
10. The connection module according to claim 7, wherein the board holding portion includes: a holding body portion that extends along the board body member, and a retaining piece that is spaced from the holding body portion and holds the board body member with the holding body portion.
11. The connection module according to claim 7, wherein the board body member includes a positioning hole, and the board holding portion includes a holding body portion that extends along the board body member and a positioning projection that projects from the holding body portion and is inserted in the positioning hole to position the board body member with respect to the holding body portion.
12. The connection module according to claim 7, wherein each of the connection member holding portions includes a base portion that is disposed along a corresponding one of the connection members and a stopper that is continuous from the base portion, and the stopper includes a warping portion that projects from the base portion and can be warped and a stopper projection that projects from the warping portion and sandwiches and holds the corresponding one of the connection members with the base portion.
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